

Exp No: 14

EMBEDDED C PROGRAM TO DISPLAY "HELLO WORLD" MESSAGE

The screenshot displays an IDE with the following components:

- Registers Window:** Shows the state of various registers. The 'Sys' section includes 'a' (0x00), 'b' (0x01), 'sp' (0x21), 'sp_max' (0x27), 'dptr' (0x0c39), 'PC' (0x0c27), 'status' (130691), 'pcr' (0.06534550), and 'psw' (0x00).
- Code Editor:** Contains the following C code:

```
1 #include <reg51.h>
2 #include <stdio.h>
3
4 void main (void)
5 {
6     SCON = 0x50; /* SCON: mode 1, 8-bit UART, enable rscr */
7     TMOD = 0x20; /* TMOD: timer 1, mode 2, 8-bit reload */
8     TH1 = 0x00; /* TH1: reload value for 9600 baudrate */
9     TL1 = 1; /* TL1: timer 1 run */
10    TI = 1; /* TI: set TI to send first char of UART */
11    while (1)
12    {
13        printf ("Hello World ! \n ");
14    }
15 }
```
- Command Window:** Shows the execution status: "Running with Code Size Limit: 2K" and "Load 'C:\Users\raghu\OneDrive\Documents\8051rzh\Objects\raghunath'". It lists memory addresses and values for variables like 'sum', 'numbers[j]', 'temp', 'Hexval', 'ten', and 'hundred'.
- UART #1 Window:** Displays the output of the program, showing "Hello World !" repeated multiple times.